STUDENT RETENTION & DROPOUT ANALYTICS

EduPulse

Turning student data into actionable insights

SQL | Python | Power BI | Statistical Analysis | n8n + AI

EDUPULSE

STUDENT RECORDS	4,424
DROPOUT RATE	32.12%
GRADUATE	49.93%



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THE BUSINESS PROBLEM

Dropout is a business problem, not just an academic metric

Educational institutions need earlier visibility into academic and financial patterns associated with dropout.



What proportion of students graduate, remain enrolled, or drop out?



Which academic and financial patterns are associated with higher observed dropout rates?



How can validated findings become practical support actions?

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FROM DATA TO DECISION

One dataset, one workflow, multiple decision lenses

4,424 cleaned student records from the UCI Student Dropout and Academic Success dataset

01

Data

UCI Student Dropout and Academic Success dataset

4,424 cleaned student records

02

Analysis

SQL

SQL business analysis answers the operational questions.

03

Validation

Python · Pandas · NumPy · SciPy

EDA, data quality, and statistical validation.

04

Dashboard

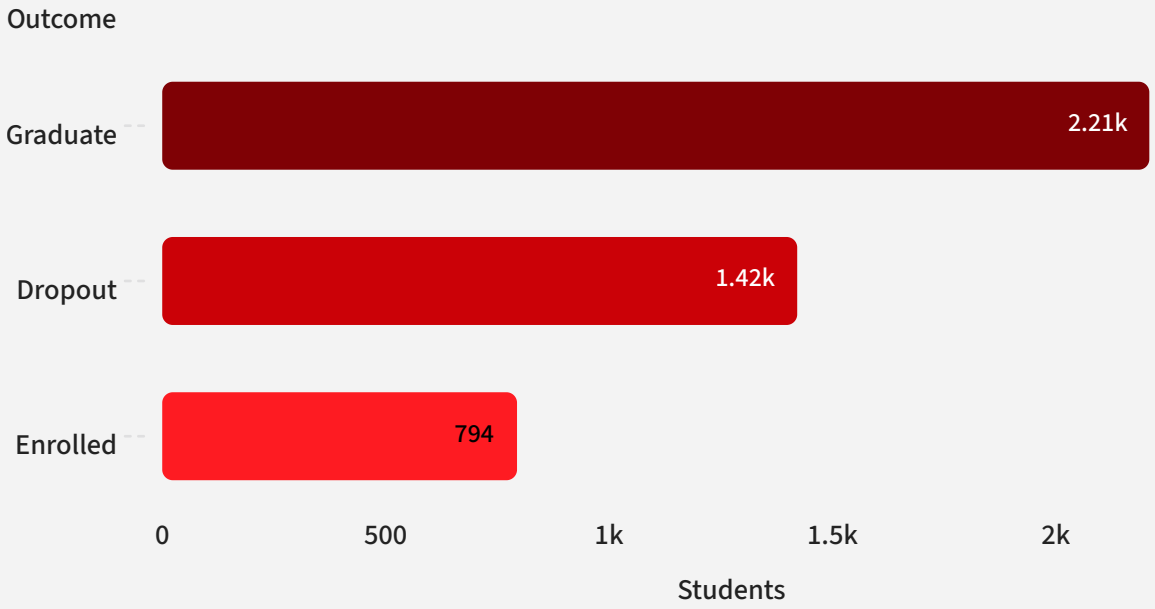
Power BI

Turns the findings into an interactive three-page dashboard.

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OUTCOME DISTRIBUTION

Half graduate, one-third dropout, one in six still enrolled



Outcome	Students	Share
Graduate	2,209	49.93%
Dropout	1,421	32.12%
Enrolled	794	17.95%

Total population: 4,424 students

The observed population is led by graduates, with 1,421 students recorded as dropout.

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KEY FINDINGS

Academic progress and financial friction separate risk groups

Lower academic progress and financial friction appear together in the highest-risk groups.

Academic progress

Driver	Lower-risk group	Higher-risk group
2nd-semester grade	12.70 graduate	**5.90 dropout**
Approved units	6.18 graduate	**1.94 dropout**

Financial friction

Driver	Lower-risk group	Higher-risk group
Tuition fee status	24.7% up to date	**86.5% not up to date**
Debtor status	28.3% non-debtor	**62.0% debtor**
Scholarship status	12.2% holders	**38.7% non-holde**

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COMBINED SEGMENT

The clearest risk signal is a combined segment

Observed dropout rate

Academic progress	Fees up to date	Overdue tuition fees
Grade 8 or above	—	—
Grade below 8	—	**97.0%** observed dropout rate

STATISTICAL VALIDATION

Welch's t-tests, Chi-square tests, and Cramér's V examined differences and associations.

Tuition fee status

0.431

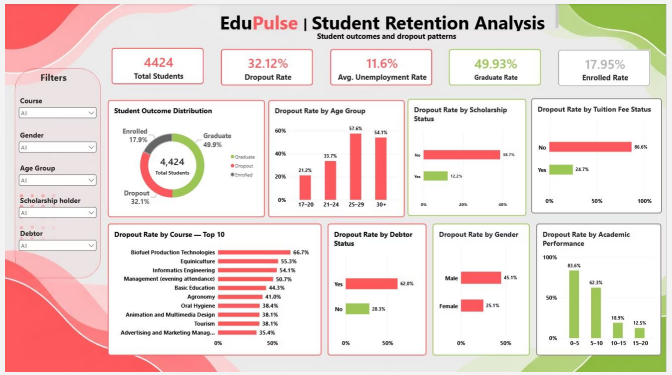
Cramér's V, strongest association among the three highlighted financial indicators.

Interpretation: association, not causation; the analysis does not predict an individual's future outcome.

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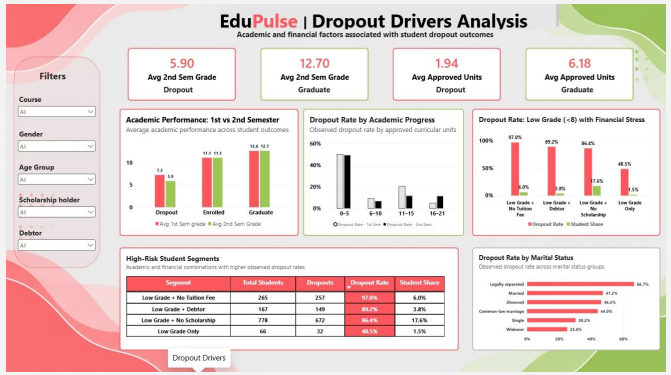
POWER BI WALKTHROUGH

The dashboard turns analysis into a decision surface



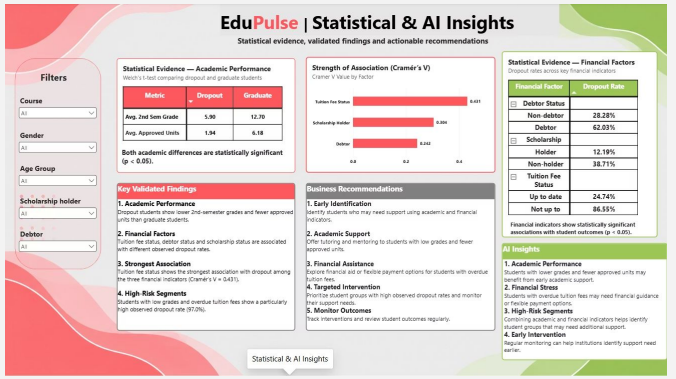
Executive Overview

Outcomes, dropout rate, age groups, and financial indicators.



Dropout Drivers

Academic performance, approved curricular units, and high-risk segments.



Statistical & AI Insights

Statistical evidence, Cramér's V, and business recommendations.

Strongest signal

Tuition fee status · Cramér's V = 0.431

Decision surface

From validated findings to business recommendations.

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RECOMMENDATIONS

From validated analysis to action, automatically

1

SHARED STAGE

Python

Prepares data, checks quality, and produces validated results.

2

AI Business Report → n8n → AI → Gmail

n8n receives the results through a webhook and routes the business-report path. AI interprets the evidence into business-friendly insights. Gmail delivers the AI report when the configured condition is met.

3

Technical Data Quality Alert → notifier.py → Gmail

Issues are detected in the technical data quality path. Python SMTP sends technical anomaly alerts when issues are detected.

Recommendations

01

Early identification

02

Academic support for low grades and fewer approved units

03

Financial assistance or flexible payment options

04

Targeted intervention and outcome monitoring

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EDUPULSE

Thank You

EduPulse turns student data into earlier, more targeted support.

Neeraj Singh · Data Analytics | SQL | Python | Statistics | Power BI | AI Automation

Next step: connect interventions to outcomes and evaluate impact.



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